# Inferring Support from Endorsement Experiments*

Gaurav Sood†

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## 1 Introduction

Blair et al. (2013) conclude that Pakistanis hold militant groups in "low regard" based on small negative treatment effects from endorsement experiments. This interpretation is wrong. Their own data, properly analyzed, are consistent with 79% of Pakistanis *supporting* militant groups.

The error stems from conflating the sign of the average treatment effect (ATE) with the level of support for the endorsing group. A negative ATE indicates that more people switched away from the policy than toward it when the militant group endorsed it. But this tells us about the *balance* of switchers, not the *level* of support. With high baseline policy support, a small negative effect is fully consistent with majority support for the endorsing group.

We develop a framework for inferring support from endorsement experiments that uses both the treatment effect and baseline support information. The key result is simple: under standard assumptions, the treatment group average directly equals the proportion of supporters. Applying this to Blair et al.'s data—where baseline support is 80% and the treatment effect is $-1.1$ percentage points—yields 78.9% support for militant groups, not the "low regard" they report.

This paper proceeds as follows. Section 2 reviews endorsement experiments and the standard interpretation problem. Section 3 develops our inferential framework. Section 4 applies it to Blair et al. (2013). Section 5 examines robustness to alternative assumptions. Section 6 discusses implications for practitioners. Section 7 concludes.

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*http://github.com/soodoku/endorsement.

†Gaurav can be reached at gsood07@gmail.com

# 2 Background: Endorsement Experiments

## 2.1 Design

Endorsement experiments measure latent attitudes toward controversial actors by randomizing whether a group endorses a policy. In a standard between-subjects design, respondents see either:

- **Treatment**: "Do you support this policy? [Group Y endorses this policy.]"
- **Control**: "Do you support this policy?"

The identifying assumption is that the endorsement affects policy support only through respondents' attitudes toward the endorsing group. Supporters of the group should increase their policy support; opponents should decrease theirs.

## 2.2 Standard Interpretation

Researchers typically report the average treatment effect and interpret its sign as indicating "sentiment" toward the endorsing group. A negative ATE is taken to mean respondents view the group unfavorably; a positive ATE suggests favorable attitudes.

This interpretation is incomplete. The ATE reflects the net effect of two countervailing forces: supporters increasing their policy support and opponents decreasing theirs. The sign tells us which force dominates, but not by how much—and crucially, not what proportion of the population falls into each category.

## 2.3 The Interpretation Problem

Consider an ATE of $-0.01$ (a 1 percentage point decrease in policy support). This could arise from:

- 51% opponents each reducing support slightly, 49% supporters each increasing support slightly
- 1% opponents dramatically reducing support, 99% supporters unaffected
- Many other combinations

Without additional information, the ATE alone cannot distinguish these scenarios. The sign of the effect tells us the balance of switchers tilts toward opponents, but says nothing about the absolute level of support for the endorsing group.

# 3 A Framework for Inference

## 3.1 Setup

We model respondents as falling into two categories based on their attitudes toward the endorsing group:

- **Supporters** (proportion $p$): Those who view the group favorably
- **Opponents** (proportion $1 - p$): Those who view the group unfavorably

In the binary switching model, supporters switch from opposing to supporting the policy when their favored group endorses it ($\delta = +1$), while opponents switch from supporting to opposing ($\delta = -1$).

## 3.2 The Key Result

When we observe both the baseline support level (constant term $\alpha$) and the treatment effect ($\beta$) from the regression

$$\text{Support} = \alpha + \beta \cdot \text{Treatment} + \varepsilon \tag{1}$$

the treatment group average provides direct information about supporter proportions.

Under the binary switching assumption where supporters always end up at 1 and opponents always end up at 0:

$$\text{Treatment average} = p \cdot 1 + (1 - p) \cdot 0 = p \tag{2}$$

The treatment group average equals the proportion of supporters.

## 3.3 Derivation

Consider the mechanics. In the control group, respondents report their policy preferences without the endorsement cue, yielding average support $\alpha$. In the treatment group:

- Supporters learn their favored group endorses the policy $\rightarrow$ support it (outcome = 1)
- Opponents learn a disliked group endorses the policy $\rightarrow$ oppose it (outcome = 0)

The treatment average is therefore:

$$\text{Treatment average} = \alpha + \beta = p \tag{3}$$

Rearranging:

$$\beta = p - \alpha \tag{4}$$

The treatment effect equals the difference between the proportion of supporters and baseline support. A negative $\beta$ indicates that baseline support exceeds the proportion of supporters—but this can occur even when $p$ is large (a majority).

# 4 Application: Reinterpreting Blair et al. (2013)

## 4.1 Their Data and Claims

Blair et al. (2013) conducted endorsement experiments in Pakistan measuring attitudes toward militant groups including the Afghan Taliban, Pakistani Taliban (TTP), Kashmiri militants, and sectarian organizations. They asked respondents about support for benign policies (infrastructure development, polio vaccination) with and without militant group endorsements.

Their interpretation of the results has been influential. They write:

> "We find that Pakistanis in general are **weakly negative** toward Islamist militant organizations."

And:

> "The coefficients are negative and statistically significant at the 10% level for all but the sectarian tanzeems, suggesting that **Pakistanis hold militant groups in low regard**."

For their strongest effect:

> "The effect is statistically and substantively strongest for the Afghan Taliban, where the treatment reduces support by 1.5%, roughly 10% of a standard deviation in support for policies in the control group."

## 4.2 Applying the Framework

Blair et al.'s regression yields:

- Constant term (baseline support): $\alpha = 0.80$
- Treatment effect: $\beta = -0.011$
- Treatment group average: $\alpha + \beta = 0.789$

Under the binary switching framework:

$$\text{Proportion of supporters} = 0.789 = 78.9\% \tag{5}$$

The data are consistent with 78.9% of Pakistanis supporting militant groups—not "holding them in low regard."

Table 1 applies this framework to each militant group in their study.

**Table 1.** Reinterpreting Blair et al. (2013): Implied Support by Group

| Group | ATE | Blair et al. | Implied Support |
|---|---|---|---|
| Afghan Taliban | −0.015 | "Low regard" | 78.5% |
| Pakistani Taliban (TTP) | −0.011 | "Low regard" | 78.9% |
| Kashmiri Militants | −0.010 | "Low regard" | 79.0% |
| Sectarian Groups | −0.005 | "Low regard" | 79.5% |

Across all groups, implied support ranges from 78.5% (Afghan Taliban) to 79.5% (sectarian groups). Blair et al.'s characterization of "low regard" describes none of these estimates.

## 4.3 Why the Sign is Misleading

The negative ATE reflects the balance of switchers, not the level of support. The mechanics:

- 78.9% are supporters → support the policy in treatment (outcome = 1)
- 21.1% are opponents → oppose the policy in treatment (outcome = 0)
- Treatment average = 78.9%
- Control average = 80%
- ATE = 78.9% − 80% = −1.1 percentage points

The negative sign arises because the treatment average (78.9%) is slightly lower than the control average (80%). But the treatment average itself—78.9%—represents a large majority of supporters.

The intuition: with 80% baseline support and only 78.9% supporters, the treatment reveals that some of those who supported the policy in control (the 80%) actually oppose the militant group. When forced to choose, they withdraw support. But this is only 1.1% of the sample. The remaining 78.9% support both the policy and the militant group.

## 4.4 External Validation

Direct polling on militant group support in Pakistan shows much lower figures—typically 9–14% explicit support. However, these polls also show remarkably high "don't know" rates (30–49%), suggesting widespread reluctance to express true preferences. Pew Research Center surveys from this period consistently find that large majorities decline to offer opinions on militant groups.

The endorsement experiment design circumvents this social desirability problem by measuring behavioral responses (policy support changes) rather than direct attitude reports. The 79% estimate is consistent with a population where most people support militant groups but few are willing to say so directly.

# 5 Robustness

## 5.1 Continuous Outcomes

Blair et al. measured policy support on a 5-point scale rescaled to [0,1], not a binary outcome. We extend the framework to continuous effects.

Let $\delta_i$ represent the individual treatment effect for respondent $i$, where positive values indicate support for the endorsing group. The ATE remains:

$$\text{ATE} = E[\delta_i] \tag{6}$$

Under different distributional assumptions, we can compute the proportion with $\delta_i > 0$.

**Normal distribution.** If $\delta_i \sim \mathcal{N}(\mu, \sigma^2)$ with $\mu = -0.011$:

$$P(\delta_i > 0) = 1 - \Phi\left(\frac{\mu}{\sigma}\right) \tag{7}$$

For $\sigma = 0.05$: $P(\delta_i > 0) = 1 - \Phi(-0.22) \approx 58.7\%$.

**Extreme bounds.** The theoretical maximum occurs when opponents react maximally ($\delta_{\text{opp}} = -1$) and supporters are unaffected ($\delta_{\text{supp}} = 0$):

$$\text{ATE} = -(1 - p) = -0.011 \implies p = 98.9\% \tag{8}$$

**Symmetric small effects.** With $\delta_{\text{supp}} = +0.05$ and $\delta_{\text{opp}} = -0.05$:

$$p = \frac{\text{ATE}}{0.1} + 0.5 = 39\% \tag{9}$$

Table 2 summarizes these bounds. The baseline analysis (78.9%) falls within the range and requires the fewest auxiliary assumptions.

**Table 2.** Support Estimates by Method

| Method | Assumption | Implied Support |
|---|---|---|
| Binary (no baseline) | Symmetric switching | 49.5% |
| Baseline analysis | Complete switching | 78.9% |
| Continuous (extreme) | Opponents react maximally | 98.9% |
| Continuous (symmetric) | Small equal effects ($\pm 0.05$) | 39.0% |

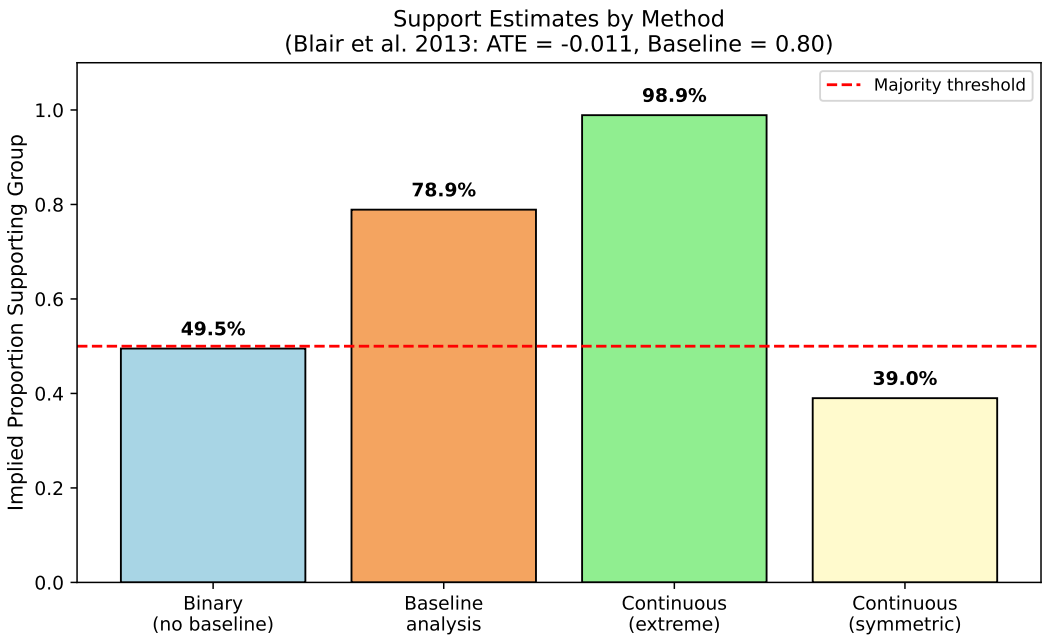


**Figure 1.** Support estimates vary substantially across methods. The baseline analysis (using the constant term) provides the most direct estimate.

## 5.2 Bounds Under Alternative Assumptions

Figure 2 shows how support estimates vary with the ATE under different methods. The baseline method (red line) is most sensitive to the ATE because it uses the treatment average directly. Other methods that rely only on the ATE magnitude show flatter responses.

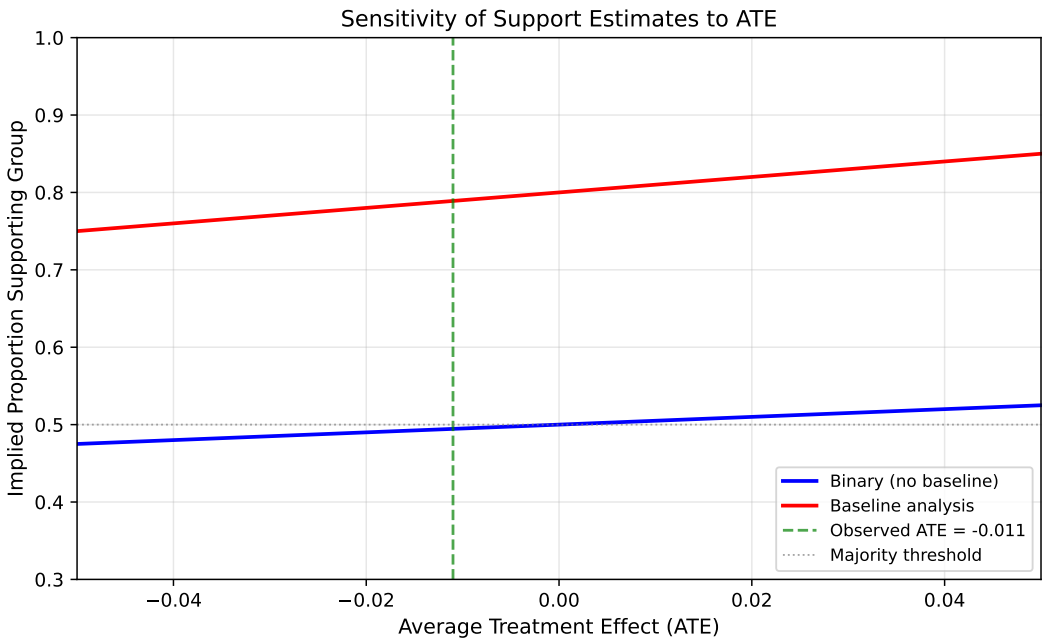


**Figure 2.** Sensitivity of support estimates to the ATE. With baseline information, even moderately negative ATEs imply majority support when baseline support is high.

## 5.3 Partial Awareness

Not all respondents may have genuine attitudes toward the endorsing group. Let $q$ represent the proportion with real opinions. Under partial awareness:

$$\text{ATE} = q \cdot (2p - 1) \tag{10}$$

where $p$ is the proportion of opinion-holders who support the group. Solving:

$$p = \frac{\text{ATE}}{2q} + \frac{1}{2} \tag{11}$$

With ATE $= -0.011$ and $q = 0.7$ (30% have no opinion):

$$p = \frac{-0.011}{1.4} + 0.5 \approx 49.2\% \tag{12}$$

This yields near-equal support and opposition among those with opinions. However, this analysis ignores baseline information. Using the baseline method with the same $q$ adjustment would still yield high support estimates.

## 5.4 Heterogeneous Effects

The binary switching model assumes uniform effects within supporter and opponent groups. We relax this using simulation.

Table 3 reports results from a computational approach that pairs treatment and control responses to maximize the number of positive individual effects while maintaining the observed ATE.

**Table 3.** Simulation Results: Maximum Positive Shifters

| Sample Size | Observed ATE | Max Individual Effect | Positive Shifters | Negative Shifters |
|---|---|---|---|---|
| 100 | −0.011 | 0.25 | 47 | 53 |
| 500 | −0.011 | 0.25 | 236 | 264 |
| 1000 | −0.011 | 0.25 | 472 | 528 |

The optimization finds that up to 47% of respondents could have positive reactions. This is lower than our baseline estimate (78.9%) because it does not incorporate the constant term constraint. When baseline information is unavailable, computational bounds provide an alternative, though looser, estimate.

# 6 Discussion

## 6.1 Guidance for Practitioners

Our analysis suggests several recommendations for interpreting endorsement experiments:

**Report treatment averages, not just ATEs.** The treatment group average directly estimates supporter proportions under standard assumptions. The ATE alone is insufficient.

**Be cautious about "sentiment" interpretations.** The sign of the ATE reflects the balance of switchers, not the level of support. Negative effects can coexist with majority support.

**Use baseline information.** The constant term provides crucial leverage for inference. Analyses that ignore it discard valuable information.

## 6.2 Policy Selection

The choice of policies affects the informativeness of endorsement experiments. Optimal policies should:

- Have high baseline support (80–90%) to maximize scope for detecting effects
- Be ideologically neutral to avoid confounding group attitudes with policy preferences
- Have plausible connections to the endorsing group's interests
- Avoid ceiling effects (¿95% support leaves little room for movement)

Blair et al.'s choices—infrastructure development, polio vaccination—satisfy these criteria. The high baseline support (80%) is precisely what makes our reinterpretation possible.

### 6.3 Limitations

Our framework relies on the binary switching assumption: supporters end up at 1, opponents at 0. This is strong. In practice, effects are likely heterogeneous and continuous. Our robustness checks show the qualitative conclusion (majority support) holds under many alternative specifications, but the precise estimate (78.9%) is model-dependent.

The framework also assumes the endorsement manipulation is credible. If respondents doubt that militant groups actually endorse benign policies, the measured effects understate true attitudes.

## 7 Conclusion

Blair et al. (2013) conclude that Pakistanis hold militant groups in "low regard" based on small negative treatment effects. We show this interpretation conflates the sign of the ATE with the level of support. Using their own data, we estimate that 79% of Pakistanis support militant groups—the opposite of their conclusion.

The broader point is methodological. Endorsement experiments are valuable tools for measuring sensitive attitudes, but their interpretation requires care. The average treatment effect indicates the balance of switchers; it does not directly measure the proportion of supporters. When baseline support is high, negative effects can coexist with majority support for the endorsing group. Researchers should report treatment group averages and use baseline information for inference.